

CEL: Comprehensive Counterfactual Explanations Library and Benchmark

Oleksii Furman · Łukasz Lenkiewicz · Marcel Musiałek · Maciej Zięba



genwro.AI



Benchmark scope

18 Datasets

13 classification
5 regression

14 Methods

10 Local
2 Global
2 Group-wise

2 Backbones / Task

Classification: LR + MLP
Regression: Linear + MLP

9 Counterfactual Explanations Metrics

Validity · Coverage · Sparsity
Proximity · Runtime
Prob. plausibility · Log-density
LOF · Isolation Forest

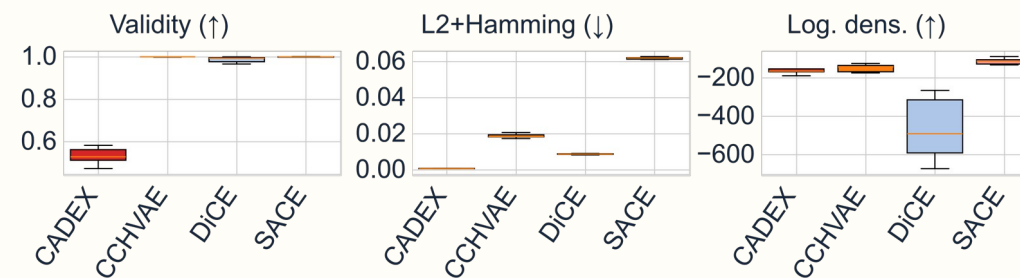
Bring your method to the benchmark.



Results

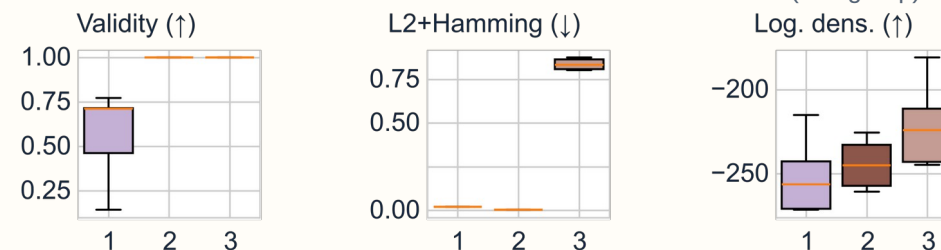
Local

No single winner



Global

Validity and failures



Group-wise

Reach versus change

